AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 29 · STUDENT LESSON

Performance and cost experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Keep every duration. Keep quality visible. State the trade-off.

Compare supplied speed and cost observations with explicit workload, mean-budget and quality limits.

Keep one habit: Measurement contract → mean/median ledger → six-case cost picture → failure probes → arithmetic audit → independent C/D case.

MAIN LESSON · SECTION 1

Compare time, quality and cost on the same workload

> Keep slow cases and failed outcomes visible.

Learning goals

I can explain the **mechanism** and terminology of performance and cost experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve denominators and mandatory quality conditions

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Accessibility and user feedback. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
>
>
>
>

End goal

Explain performance and cost experiments using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of performance and cost experiments

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Declare workload, timing boundary and quality requirements

> A speed comparison without its measurement contract can mislead.

Begin with one unusually slow question

Imagine two school helpers answer the same four questions. One is usually quick but takes nine seconds on one question; the other always takes two seconds. The mean describes the total time divided by four. The median describes the middle of the sorted times. Neither replaces the individual record, quality results or the budget. Declare where timing starts and ends before interpreting an experiment.

Define matched performance experiments

Performance experiments compare **matched workload**s with declared timing boundaries and quality constraints. Averages can hide slow cases.

Complete timing and cost contract

Experiment item	Complete supplied-fixture rule
Matched main workload	Ordered fictional cases Q1–Q4; same environment and rubric assumed
Timing boundary	Supplied seconds from request start to final outcome; includes failed outcomes in added probe

Experiment item	Complete supplied-fixture rule
Main A durations	[1, 1, 1, 9] seconds; 4/4 pass
Main B durations	[2, 2, 2, 2] seconds; 4/4 pass
Mean	Sum of all recorded durations / number of recorded cases
Median with four cases	Sort; average positions 2 and 3
Tail disclosure	Show every duration and observed maximum; four cases do not establish a population tail percentile
Separate picture	Six questions each; A total 18 seconds/12 units, B 12 seconds/18 units; supplied equivalent quality
Changed picture budget	Mean cost at most 2 fictional units per case; totals cannot establish every individual request cost
Limits	Constructed observations, no actual benchmark, repeated runs, prices or universal winner

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace mean, median and the slow recorded case

> Show the middle-pair calculation rather than choosing the most flattering statistic.

Original matched-case and median procedure

Use the same cases and environment; record every duration and success. For a simple **median** sort values and average the middle pair for even counts. Keep failed requests visible.

Complete per-configuration calculation ledger

Main supplied measure	A: [1, 1, 1, 9] seconds	B: [2, 2, 2, 2] seconds
Total / count	12 / 4	8 / 4
Mean seconds	3	2
Middle pair after sorting	1 and 1	2 and 2
Median seconds	1	2
Observed maximum seconds	9	2
Quality on these cases	4/4 pass	4/4 pass
Scoped comparison	Lower median; one slow 9-second case	Lower mean and observed maximum; no universal preference without objective

State which objective the comparison serves

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Performance and cost experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and median.	Use the same cases and environment; record every duration and success. For a simple median sort values and average the middle pair for even counts. Keep failed requests visible.	A mean 3, median 1; B mean 2, median 2. B lowers mean but A has lower median; show the 9-second tail instead of declaring one universal winner.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict a mean-cost budget while holding quality fixed

> A faster option may fail the mandatory resource rule.

Current task and response

- Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed average-cost condition

- Change: Require at most 2 cost units per case on average.

▼ **Check after predicting**

Prediction: A meets the budget; B does not.

Reason: The mandatory resource condition changes the decision.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the time/cost trade-off under an explicit average budget

> The denominator and the resource rule determine the scoped conclusion.

Main four-case means/medians and checked conclusion

Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**.

A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate six-case aggregates and complete calculations

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

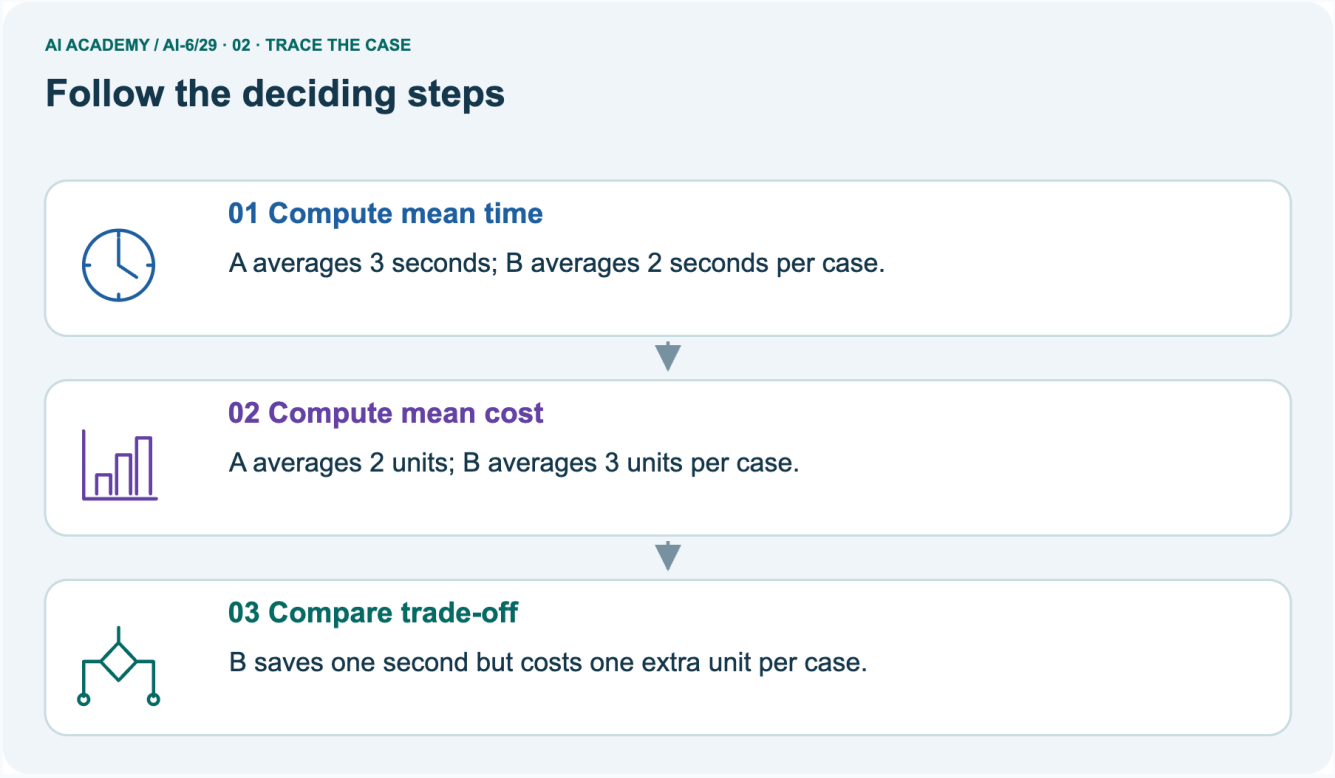
A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

The picture is a separate six-question aggregate, not a continuation of the main four durations. A averages 18/6 = 3 seconds and 12/6 = 2 cost units; B averages 12/6 = 2 seconds and 18/6 = 3 units. B saves one second per case on average while using one extra cost unit. Under a mean-cost budget of at most 2 units, A meets equality and B fails.

Individual request costs and durations are not supplied for this picture, so do not claim an individual cap, median or maximum. Cost units are fictional, not money. Quality is supplied as meeting the same rubric; broader reliability and repeated-run behavior remain unmeasured.

Separate picture: six cases each	**A**	**B**
Total seconds / cases	18 / 6	12 / 6
Mean seconds per case	3	2
Total cost units / cases	12 / 6	18 / 6
Mean cost units per case	2	3
Mean budget ≤ 2 units	Meets inclusive boundary	Fails
Not inferable from totals	Individual max, median, cost cap	Individual max, median, cost cap

Trace average time, average cost and objective



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute mean time	A averages 3 seconds; B averages 2 seconds per case.
2. Compute mean cost	A averages 2 units; B averages 3 units per case.
3. Compare trade-off	B saves one second but costs one extra unit per case.

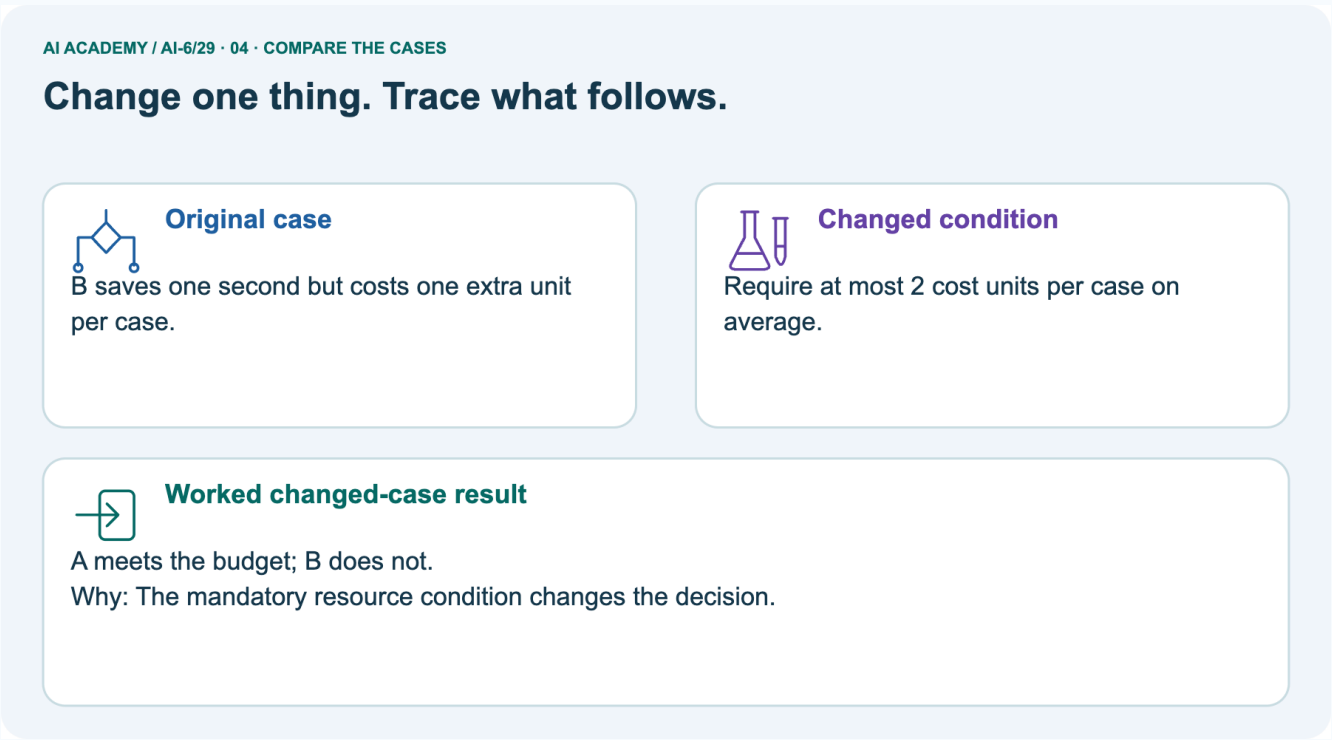
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

Predict mean budget at most 2 units

- Change one thing: Require at most 2 cost units per case on average. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain variation and missing individual records

- Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

- Read the labelled picture: Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?

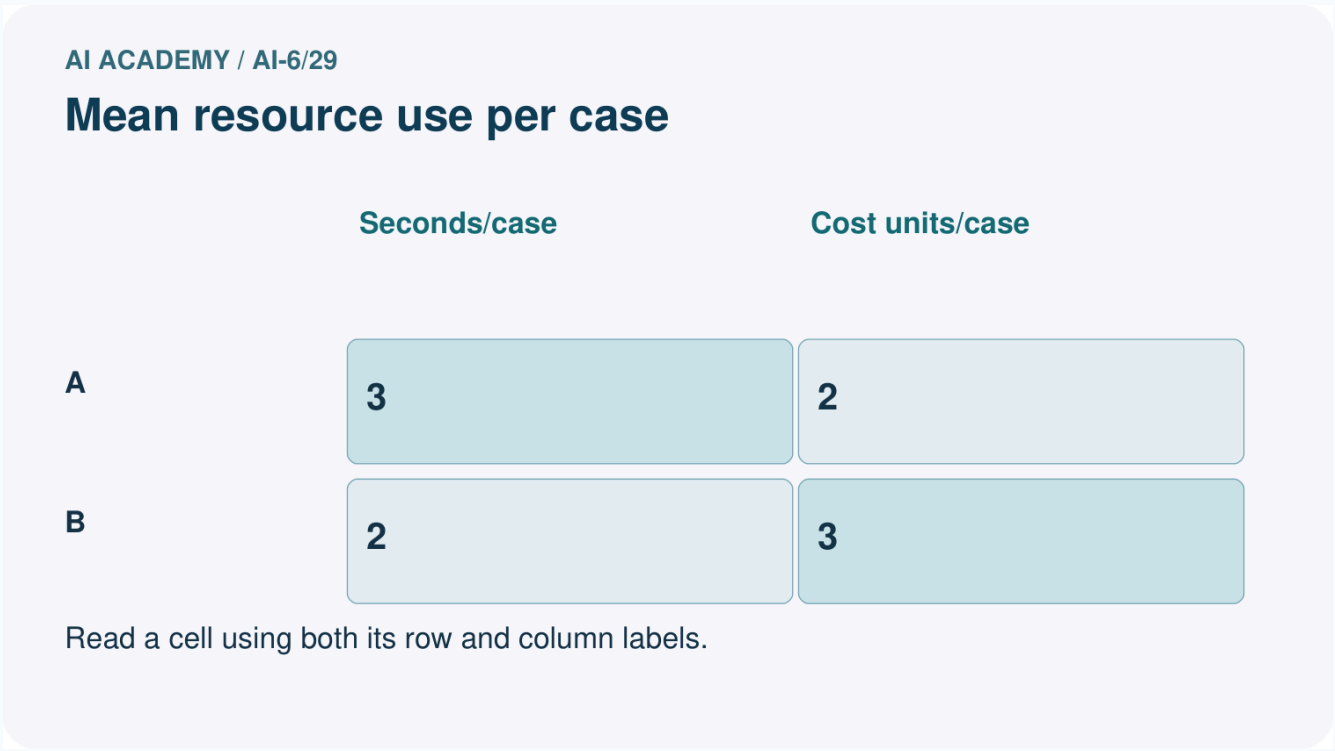
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Performance and cost experiments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/29

Performance and cost experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute mean time

A averages 3 seconds; B averages 2 seconds per case.



2. Compute mean cost

A averages 2 units; B averages 3 units per case.



3. Compare trade-off

B saves one second but costs one extra unit per case.

CHANGE ONE THING

Require at most 2 cost units per case.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit hidden failures, omitted slow cases and aggregate overclaims

> A useful comparison preserves what makes the faster option less useful.

Core performance qualification and failure analysis

Resource improvement under one workload is not universal efficiency or equal quality.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete workload, quality and cost probes

Performance probe	Expected interpretation
Mean improves while median worsens	Report both and raw durations; no universal winner
One slow case removed from report	Biased summary; preserve all supplied eligible outcomes
Fast response is wrong	Keep its duration and report quality failure; speed is not successful task completion
Mean budget met at equality	Allowed under at-most rule
Total cost/count within budget	Establishes mean only, not an individual-request maximum
Different case IDs/order	Reject paired fixture comparison until workload is matched
Negative, NaN or Boolean duration	Reject invalid supplied measurement
Four cases, no repeated trials	No stable tail behavior or precision claim

Performance probe	Expected interpretation
Latency/cost preference unclear	State the objective and mandatory constraints before choosing

Guided repair and explanation

- A report omits this qualification: Resource improvement under one workload is not universal efficiency or equal quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise explaining a comparison with its objective

> Keep the individual evidence and quality count beside the averages.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example “Performance and cost experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Performance and cost experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?

My explanation using a pictured step and the given facts

Reserve the selected C/D quality case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete time-summary and mean-budget investigation

> Check the raw duration/quality records before the summary.

Current four-case timing investigation

Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

Runnable supplied-observation audit

```
"""Summarize supplied durations and fictional costs; no benchmark or provider calls."""
from math import isfinite
from statistics import mean,median

def nonnegative(value):return type(value) in (int,float) and isfinite(value) and value>=0

def summarize(rows):
 if type(rows) is not list or not rows:raise ValueError('nonempty measured-case fixture
required')
 ids=[];durations=[];successes=[]
 for row in rows:
  if type(row) is not dict or set(row)!={'case_id','seconds','success'}:raise
ValueError('exact supplied fields required')
  if type(row['case_id']) is not str or not row['case_id'] or row['case_id'] in ids:raise
ValueError('unique case ID required')
  if not nonnegative(row['seconds']) or type(row['success']) is not bool:raise
ValueError('valid duration and success flag required')

ids.append(row['case_id']);durations.append(row['seconds']);successes.append(row['success'])
 return {'case_ids':ids,'durations':durations,'count':len(rows),'passes':sum(successes),

'mean_seconds':mean(durations),'median_seconds':median(durations),'max_seconds':max(durations)}
def compare(a,b):
 sa=summarize(a);sb=summarize(b)
 if sa['case_ids']!=sb['case_ids']:raise ValueError('same ordered case IDs required for
paired fixture')
 return sa,sb

def cost_means(count,total_seconds,total_cost):
 if type(count) is not int or count<=0 or not all(nonnegative(v) for v in
[total_seconds,total_cost]):
  raise ValueError('positive count and finite nonnegative totals required')
 return
```

```
{'count':count,'mean_seconds':total_seconds/count,'mean_cost_units':total_cost/count}
def rows(durations):return [{'case_id':'Q'+str(i+1),'seconds':s,'success':True} for i,s in
enumerate(durations)]
a,b=compare(rows([1,1,1,9]),rows([2,2,2,2]))
assert (a['mean_seconds'],a['median_seconds'],a['max_seconds'],a['passes'])==(3,1,9,4)
assert (b['mean_seconds'],b['median_seconds'],b['max_seconds'],b['passes'])==(2,2,2,4)
picture_a=cost_means(6,18,12);picture_b=cost_means(6,12,18)
mean_budget=2
assert picture_a['mean_cost_units']<=mean_budget and
picture_b['mean_cost_units']>mean_budget
# Added failure probe, separate from the reserved independent C/D case.
failed_rows=rows([2,2,2,2]);failed_rows[-1]['success']=False
failed=summarize(failed_rows)
assert failed['count']==4 and failed['passes']==3 and len(failed['durations'])==4
print('Main all-pass A:',a,'B:',b)
print('Separate picture average time/cost A:',picture_a,'B:',picture_b)
print('Mean budget at most 2 fictional units: A passes, B fails. Individual per-request cap
cannot be inferred from totals.')
print('Separate failure probe retains all four durations and reports quality:',failed)
print('Duration scope is supplied request-start-to-final-outcome fixture time, including
failed outcomes. No actual timer, repeated trials, confidence interval, currency pricing or
universal winner is provided.')
```

Expected outcomes and measurement limits

1. Predict A/B means, middle pairs, medians and maxima before running or tracing. The local code retains all supplied durations and quality flags and requires matching ordered case IDs. The separate six-case picture uses only supplied totals, so computes means but no invented individual median or maximum.

   An added failure probe keeps all four durations while reporting 3/4 quality; it does not use the reserved C/D inputs. These are arithmetic on constructed observations, not actual timed requests, repeated benchmarks or current provider costs. Save an actual run separately from predictions, or mark a paper trace “not run”.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compute C/D summaries and retain D’s wrong answer

> Speed does not erase a quality failure.

Complete selected independent task

C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save matched case IDs, timing scope and quality beside cost

> Keep failed cases and slow observations in the record.

Project task and artifact

Add a labelled performance and cost experiments **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of performance and cost experiments.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual arithmetic run or paper trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Explain the trade-off without inventing a universal winner

> Name the objective, constraints and remaining measurement gaps.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of performance and cost experiments

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Matched workload	The same task cases used for comparison
Median	The middle of ordered values under a declared convention
Tail latency	Slow outcomes hidden by typical-value summaries

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Resource improvement under one workload is not universal efficiency or equal quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Performance and cost experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and median.	Use the same cases and environment; record every duration and success. For a simple median sort values and average the middle pair for even counts. Keep failed requests visible.	A mean 3, median 1; B mean 2, median 2. B lowers mean but A has lower median; show the 9-second tail instead of declaring one universal winner.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

By the end, I can	How I will show it
Explain the mechanism and terminology of performance and cost experiments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.
- Which labelled step matches this detail? B saves one second but costs one extra unit per case.
- 1. Compute mean cost
- 2. Compute mean time
- 3. Compare trade-off
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare trade-off B saves one second but costs one extra unit per case. A fair performance experiment holds the workload and measurement procedure fixed.

Faster is not automatically preferable under a budget.

Changed-case question and optional hint

- Changed-case practice: Require at most 2 cost units per case on average. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. A meets the budget; B does not.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A meets the budget; B does not. The mandatory resource condition changes the decision.

Repair and retain the record

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Review Incident investigation (ai-6/26). Goal: Explain the **mechanism** and terminology of incident investigation. Short practice: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

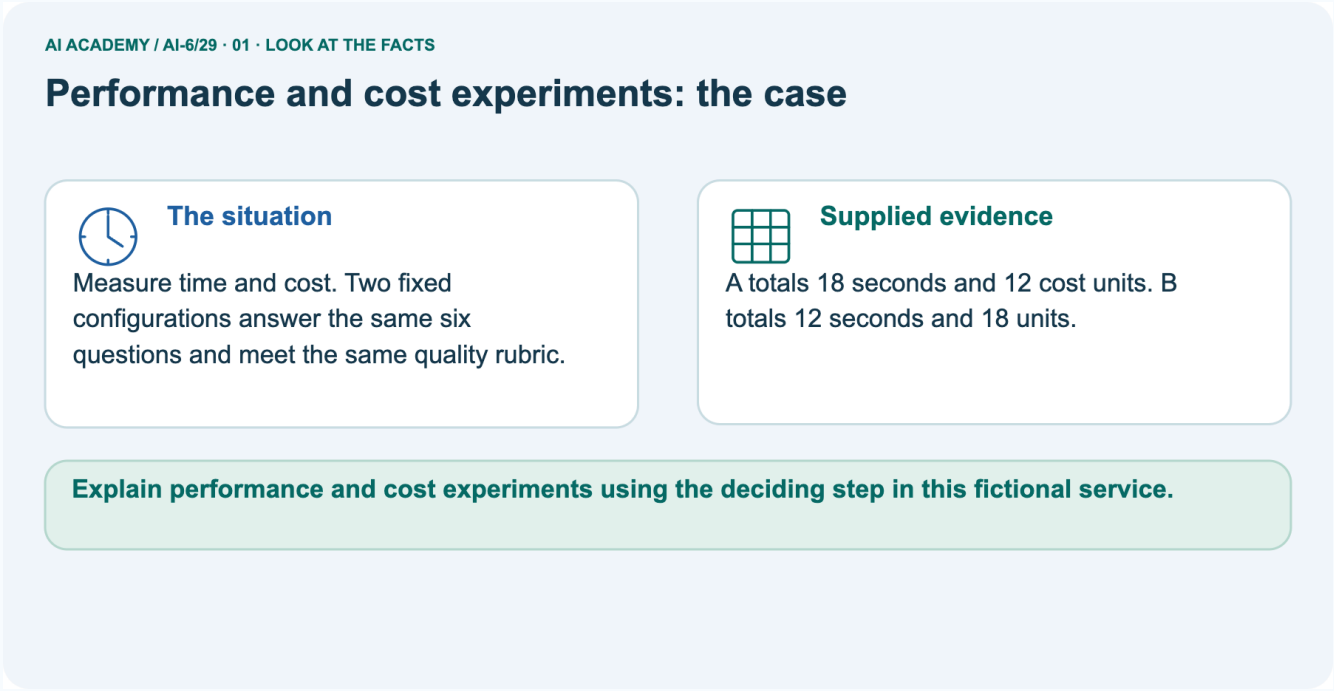
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Performance and cost experiments: the case

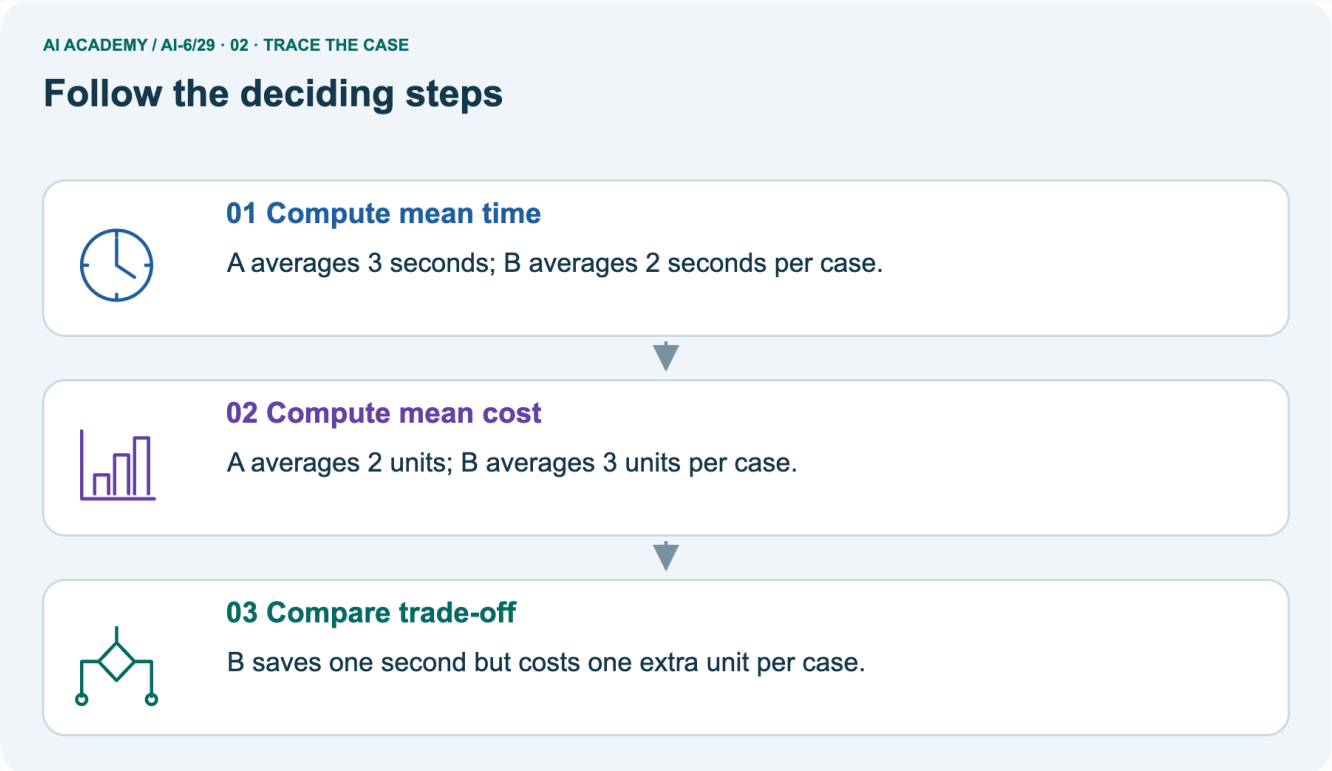


Follow the labelled picture with the matching explanation. Open the vector diagram

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric. Supplied facts: A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

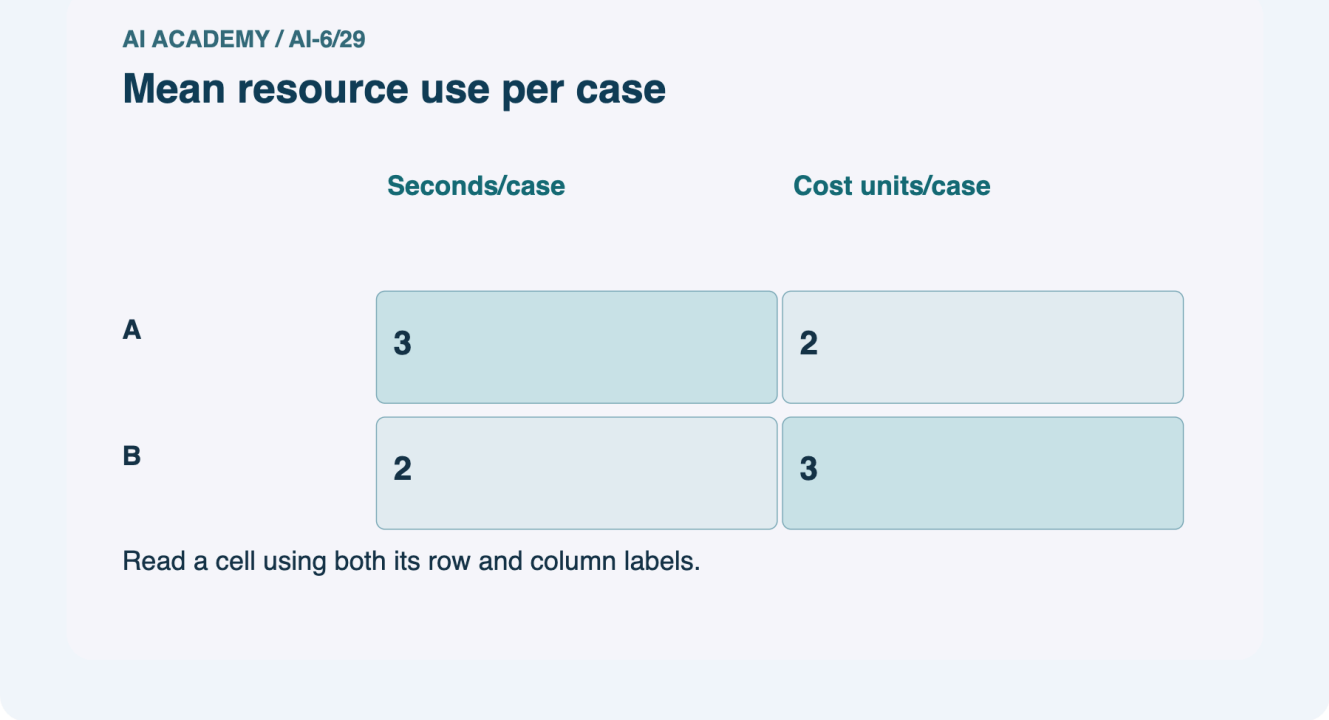


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute mean time: A averages 3 seconds; B averages 2 seconds per case. → Compute mean cost: A averages 2 units; B averages 3 units per case. → Compare trade-off: B saves one second but costs one extra unit per case.

Point to each numbered stage and explain which recorded **input** it uses.

03. Mean resource use per case

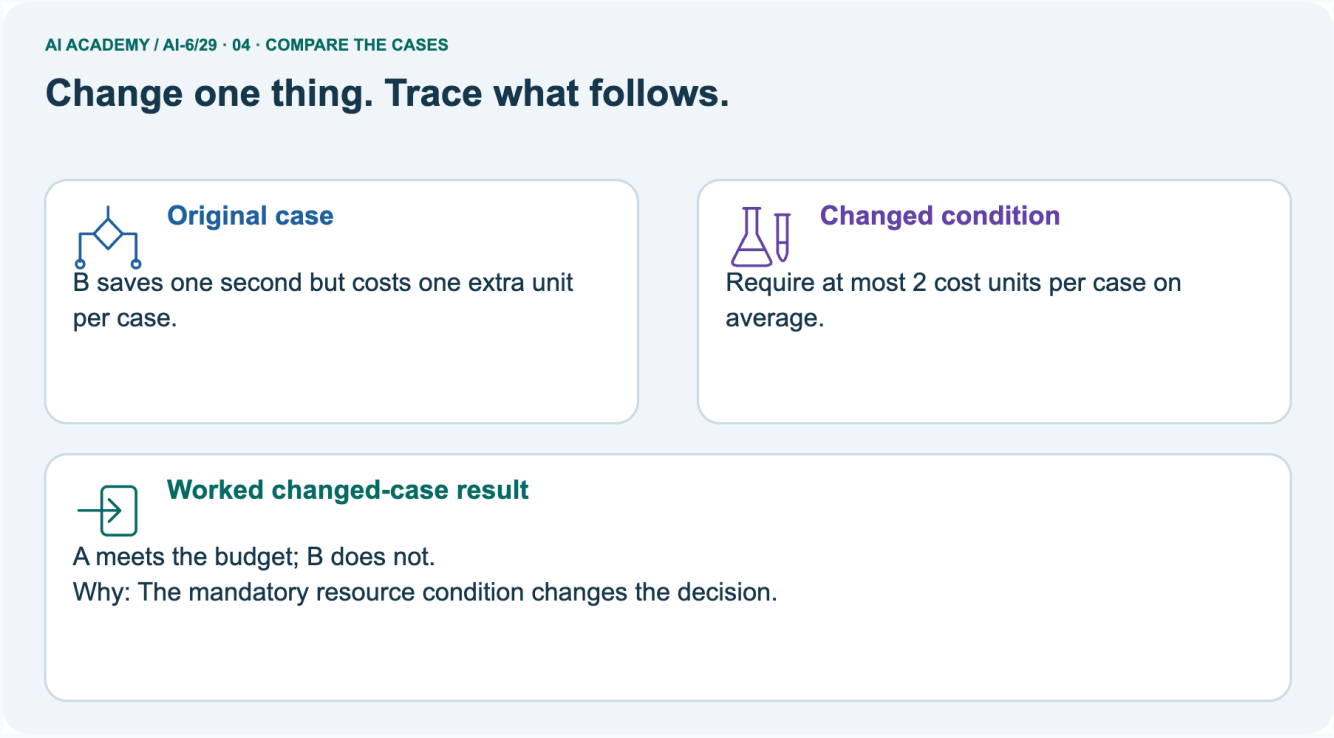


Follow the labelled picture with the matching explanation. Open the vector diagram

Mean resource use per case. A: Seconds/case = 3, Cost units/case = 2. B: Seconds/case = 2, Cost units/case = 3.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

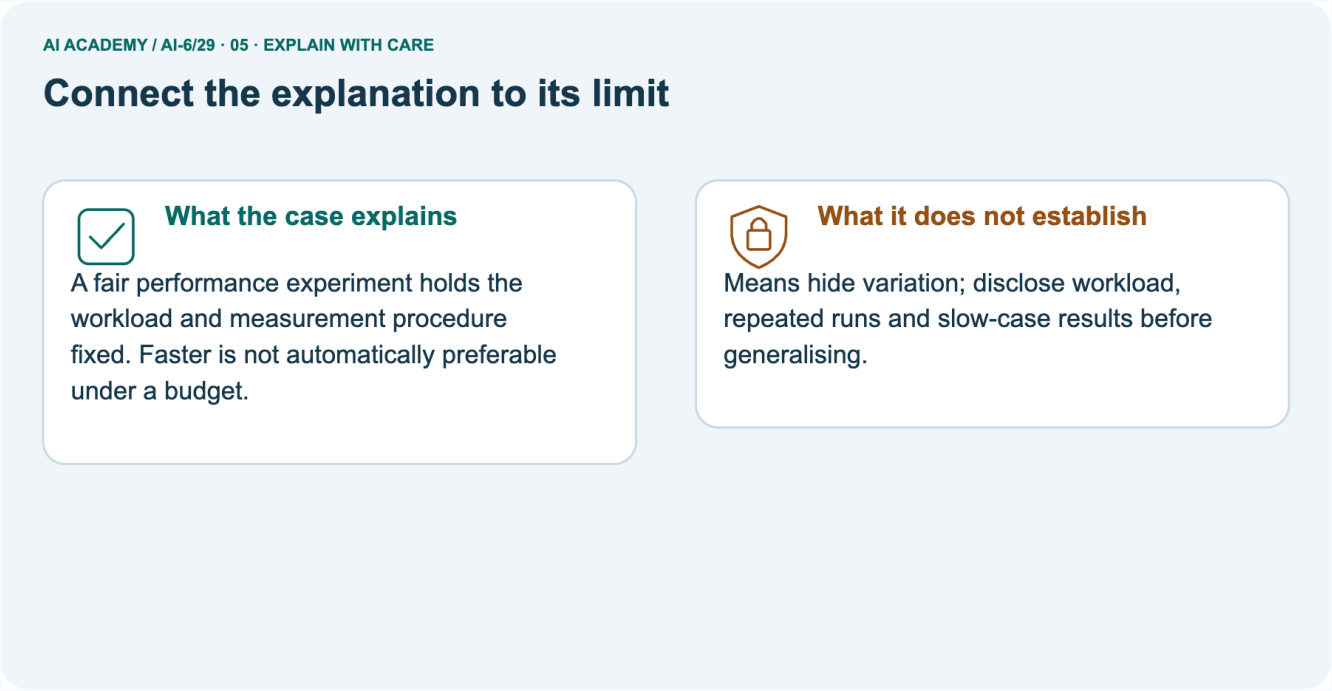


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B saves one second but costs one extra unit per case. Changed condition: Require at most 2 cost units per case on average. Expected result: A meets the budget; B does not. Reason: The mandatory resource condition changes the decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



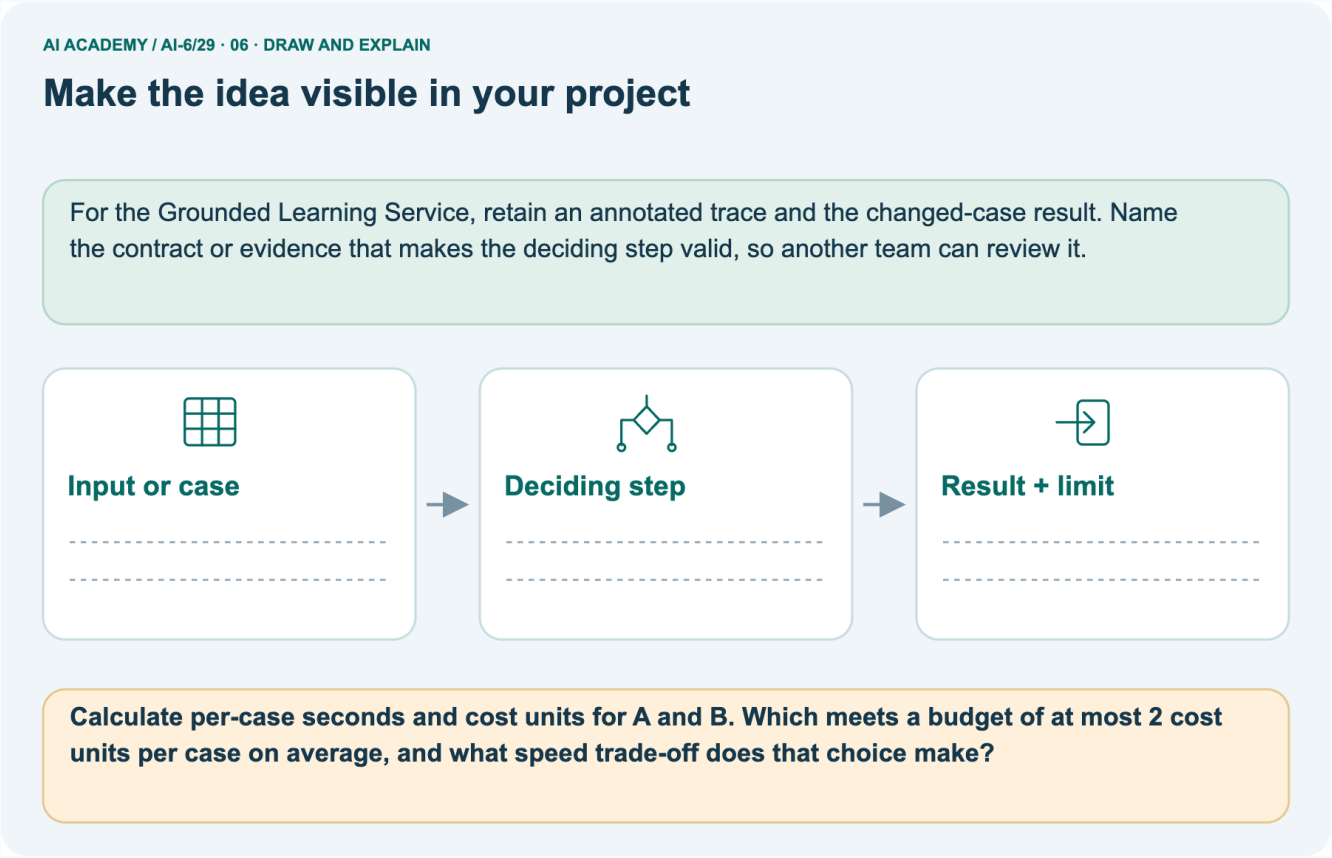
Follow the labelled picture with the matching explanation. Open the vector diagram

A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

Limit: Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it. Draw your **input**, deciding step, result and limit. Supported question: Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Performance and cost experiments: the case

02. Follow the deciding steps

03. Mean resource use per case

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.